实验二 基于灰度变换的空间域图像增强实验

1、绘制一幅灰度图像的直方图

```
I = imread('human.png');
.....
```

2、直方图均衡化对图像及其直方图的影响。

```
I=rgb2gray(imread('tree.jpg'));
.....
[m,n]=size(I);
GP=zeros(1,256);
for k=0:255
GP(k+1)=length(find(I==k))/(m*n);
end
.....
for j=1:256
for j=1:i
S1(i)=GP(j)+S1(i);
end
end
S2=round((S1*256)+0.5);
for j=1:256
GPeq(j)=sum(GP(find(S2==j)));
end
subplot(2,2,3),bar(0:255,GPeq,'b')
title('均衡化后的直方图')
xlabel('灰度值')
ylabel('出现概率')
I1=I;
```

```
for j=0:255 ,
I1(find(I==j))=S2(j+1); ,
end ,
subplot(2,2,4),imshow(I1) ,
title('均衡化后的图像'),
,
,
,
```

3、采用线性灰度变换进行图像增强处理。用 3 种映射函数（1）加常数；（2）乘常数；（3）图像灰度反转。

```
ori_img = imread('tree1.jpg'),
ori_img = rgb2gray(ori_img),
[oriHist,oriX] = imhist(ori_img),
k = 1,
d = 50,
gray2 = ori_img * k + d,
[g2Hist,g2X] = imhist(gray2),
k = 0.5,
d = 0,
gray3 = ori_img * k + d,
[g3Hist,g3X] = imhist(gray3),
k = -1,
d = 255,
ori_ = im2double(ori_img),
gray4 = ori_ * k + 1.0,
[g4Hist,g4X] = imhist(gray4),
figure(3),subplot(1,2,1),imshow(ori_img),title('原始图像'),
```

```
subplot(1,2,2),imshow(gray2),title('k=1 d=50');
figure(4),subplot(1,2,1),stem(oriX,oriHist),title('原始图像直方图');
subplot(1,2,2),stem(g2X,g2Hist),title('k=1 d=50 直方图');
figure(5),subplot(1,2,1),imshow(ori_img),title('原始图像');
subplot(1,2,2),imshow(gray3),title('k=0.5 d=0');
figure(6),subplot(1,2,1),stem(oriX,oriHist),title('原始图像直方图');
subplot(1,2,2),stem(g3X,g3Hist),title('k=0.5 d=0 直方图');
figure(7),subplot(1,2,1),imshow(ori_img),title('原始图像');
subplot(1,2,2),imshow(gray4),title('k=-1 d=255');
figure(8),subplot(1,2,1),stem(oriX,oriHist),title('原始图像直方图');
subplot(1,2,2),stem(g4X,g4Hist),title('k=-1 d=255 直方图');
```

4、采用对数变换实现图像增强

```
g=0.2*log(1+double(f));
```

5、采用伽马变换实现图像增强

```
g=double(f).^0.1;
```